

A Research Agenda for Enterprise Augmented Reality

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ABSTRACT

Over the past decade, Augmented Reality (AR) has seen significant adoption to address diverse needs in many industries including aviation, automotive, power & energy, oil & gas, manufacturing, and healthcare. However, the more practitioners leverage AR, the more industrial research focuses on domain-specific challenges. In response, the Augmented Reality for Enterprise Alliance (AREA), a program of the Object Management Group (OMG), sponsored research to develop a domain-neutral and objective, data-driven research agenda for *enterprise augmented reality*. This paper summarizes the contributions of that effort, including (i) an automated approach for gathering and hierarchically organizing state-of-the-art research about enterprise AR, (ii) a method for identifying research gaps and opportunities, and (iii) set of promising research topics from which the IEEE VGTC community could choose for future study.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

Keywords: Enterprise Augmented Reality; Industrial Augmented Reality; Research Agenda; Data-Driven Gap Analysis

1 INTRODUCTION

In many settings and use cases, enterprise AR is gaining traction. Business line leaders in large enterprises are partnering with their innovation and IT departments to evaluate AR-enabling technologies in controlled settings such as a laboratory and training center. As experience with AR increases, customers seek to expand the range of AR use cases they may evaluate, streamline time- and labor-intensive phases and increase ROI.

While AR shows promise, adoption continues to face many challenges. It is difficult for customers to develop solutions to address their specific challenges. Customers must choose to:

- invest internally to address their challenges,
- outsource custom research on specific areas to third parties,
- rely on the suppliers of commercial solutions to address the challenges for all their customers, or
- look for “seeds” of future solutions (with potentially low technology readiness) from academic centers, private research institutes or projects conducted by public/private partnerships.

A combination of these approaches is often used to reduce risk of relying on any one strategy.

2 MOTIVATION

Engineering organizations, commercial AR technology and solution providers and research groups in public and privately funded institutions who seek to address enterprise AR challenges through basic or applied research lack specific guidance from customers encountering challenges to real world adoption. Industrial research institutes or groups within commercial product companies choose research

topics based on their predicted revenue potential (ROI). The result is that many are working from the “bottom up,” rather than having clear data-driven vision guiding investments in research.

While all researchers and engineering firms would prefer to respond to “pain points” of customers, there is no objectively compiled, curated description of ecosystem needs. The lack of overarching principles and data-driven gap and needs analyses to guide researchers and funding organizations has the following impacts:

- those funding the research cannot use science and data to choose those projects most likely to result in addressing large enterprise AR customer needs and driving adoption, and
- researchers and funding agencies cannot contribute directly to one or more overarching, multi-project vision and roadmap.

3 METHODOLOGY FOR DATA-DRIVEN GAP ANALYSIS

Since there is insufficient guidance in the research community and from other public resources for how to define and prioritize future research topics, the Augmented Reality for Enterprise Alliance (AREA)¹ Research Committee developed its own methodology as part of the 8th AREA-directed Research Project.

3.1 Collecting Relevant Prior Published Works

To build a valuable enterprise AR research agenda, the relevant ongoing and past research on the topic needs to serve as the foundation. This includes both academic and industrial research. A key challenge facing this process is the lack of visibility into the landscape of industrial research activities. In other words, trends in research and development across industry are difficult to assess with minimal bias since a company’s publishing on their latest developments directly conflicts with their competitive advantage. On the other hand, academic efforts are much easier to track and to analyze since AR research results are reported through academic venues, such as journals and conference proceedings. To reduce bias and to rigorously follow a repeatable process, we only consider work published by the ACM, IEEE, and Springer communities.

3.2 Developing a System Boundary

As shown in Fig. 1-1, we leverage a well-known technical document search tool, called Engineering Village, which covers “high-quality, relevant and cross-disciplinary content, enabling engineers and researchers to perform thorough and effective literature reviews, analyze the research landscape, and solve problems” [2]. The search for past research findings includes peer-reviewed journal publications and conference proceedings but excludes standards, patents, books, and engineering dissertations. The database of work used for gap identification spans the period beginning in January 2017 and include any works available up to June 2021. If “augmented reality” appears in any metadata field, e.g., abstract, keywords, and title, the records of the item’s metadata, including but not limited to a DOI, abstract, set of keywords, publication year, and venue, related to each returned publication are stored.

3.3 Aligning Publication Indexing Services

Engineering Village leverages existing publication indexing services, most notably Inspec and Compendex. In the field of AR, Inspec and

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¹More information available at <https://thearea.org/>

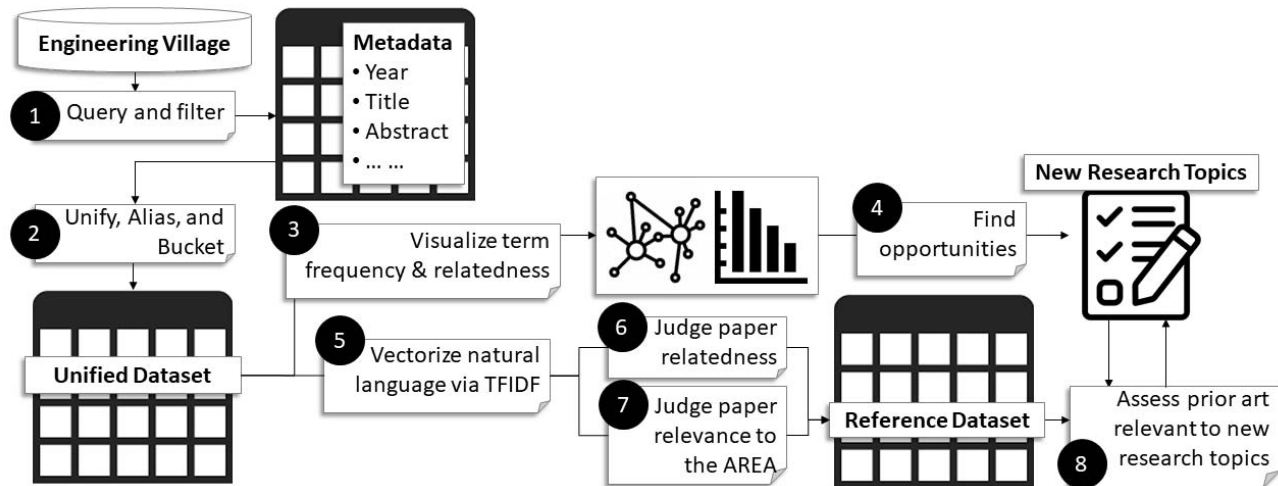


Figure 1: Summarized workflow of the methodology employed to generate new research topics for enterprise AR.

Compendex are by the far the biggest contributors to the publication database. To align these two databases, we use a data-driven methodology. Engineering Village offers three columns of keyword types, including “controlled/subject terms,” “uncontrolled terms,” and “classification codes.” The analysis performed for the AREA Research Agenda leverages “controlled/subject terms” as the ground truth keywords for the rest of the analysis, since the terms adhere to a term taxonomy defined by the two indexing services (unlike “uncontrolled terms”) yet exhibit low-level specificity to delineated critical topics (unlike “classification codes”).

As shown in Fig. 1-2, we manually created aliasing rules to (1) align terms from both indexing services with the same meaning, (2) remove British spelling and adhere to US English conventions, and (3) unify plural versions of terms with their singular versions².

3.4 Building a Hierarchy of Terms for Data Analysis

After aliasing, terms were placed within “groups” and “categories.” Groups are buckets of terms that broadly describe a research domain. There exist 61 groups, which were agreed upon by subject matter experts. For example, the Group “Emergency Responders” consists of the following terms: ambulances, disasters, emergency management, emergency services, emergency traffic control, emergency vehicles, fire protection, fires, flames, floods, and police. Note that these groupings were created through subjective means and should not be held as “ground truth” for other applications. That being said, we envision that this structured classification of terms could still be helpful for other AR R&D projects.

Categories are buckets of groups. The seven categories are Displays, End Users and User Experience, Industries, Standards, Technologies, and Use Cases. Categories were assigned using a card sorting technique. A team of professionals knowledgeable within the field of AR were presented with terms or group names and asked to sort these items into groupings based on similarity. The team then examined the percentage of agreement between all the groupings that were created. The categories contain groups of diverse size and level of detail. For example, the Category “Displays” includes wearables, display technology, optics, and device energy management. Any groupings that were not highly agreed upon were discussed until a decision for placement or the creation of a new category was finalized. The card sorts were completed using Optimal Workshop³.

²Interested readers can review the aliasing rules in our GitHub repository.

³More information can be found at <https://www.optimalworkshop.com/>.

All publication records that do not have any controlled terms after employing deletion rules are placed in the “other” group and category. Note that one-to-one mappings between terms and groups as well as groups to categories are required in this particular task. However, in practice, most terms could be assigned to multiple groups and most groups could belong in multiple categories.

3.5 Generating Frequency and Co-occurrence Charts

After terms are bucketed within group and categories, frequencies are calculated for all terms across individual records. Co-occurrences of terms across records are also calculated. Based on the frequencies and co-occurrences, a set of visualizations provided insight.

3.6 Network-based Understanding of Terms

Based on the co-occurrence values for each two-term pairing, a graph captures term-to-term relationships. In this graph, each node represents a term and each edge represents a co-occurrence. Leveraging spatial layout algorithms, we can then understand closeness of term clusters based on node and cluster proximity. We used Gephi⁴, an open graphic visualization platform, to manipulate the graphs, create sub-graphs, and encode different visual variables, including hue and color, on the nodes and edges to enhance sense-making. In doing so, we created graphs that serve as an alternative view of the dataset compared with the barcharts.

3.7 Identifying Opportunities for New Research

Based on poorly represented terms and clusters in the barcharts and graph visualizations, we identified research gaps and opportunities. Note that this was done subjectively, relying on the team’s expertise.

3.8 Vectorizing Records via NLP

Natural language, including publication titles, abstracts, and keywords, represents most of the data in the dataset. To draw insights across records, we deploy natural language processing (NLP) techniques. The foundation for these techniques is term frequency-inverse document frequency (TF-IDF). TF-IDF is a statistical measure that evaluates the importance of a word with respect to a record within a collection of records. TF-IDF takes into account the prevalence of a particular word within a record of interest and also the importance of that same word across the entirety of all the records [4].

⁴More information can be found at <https://gephi.org>

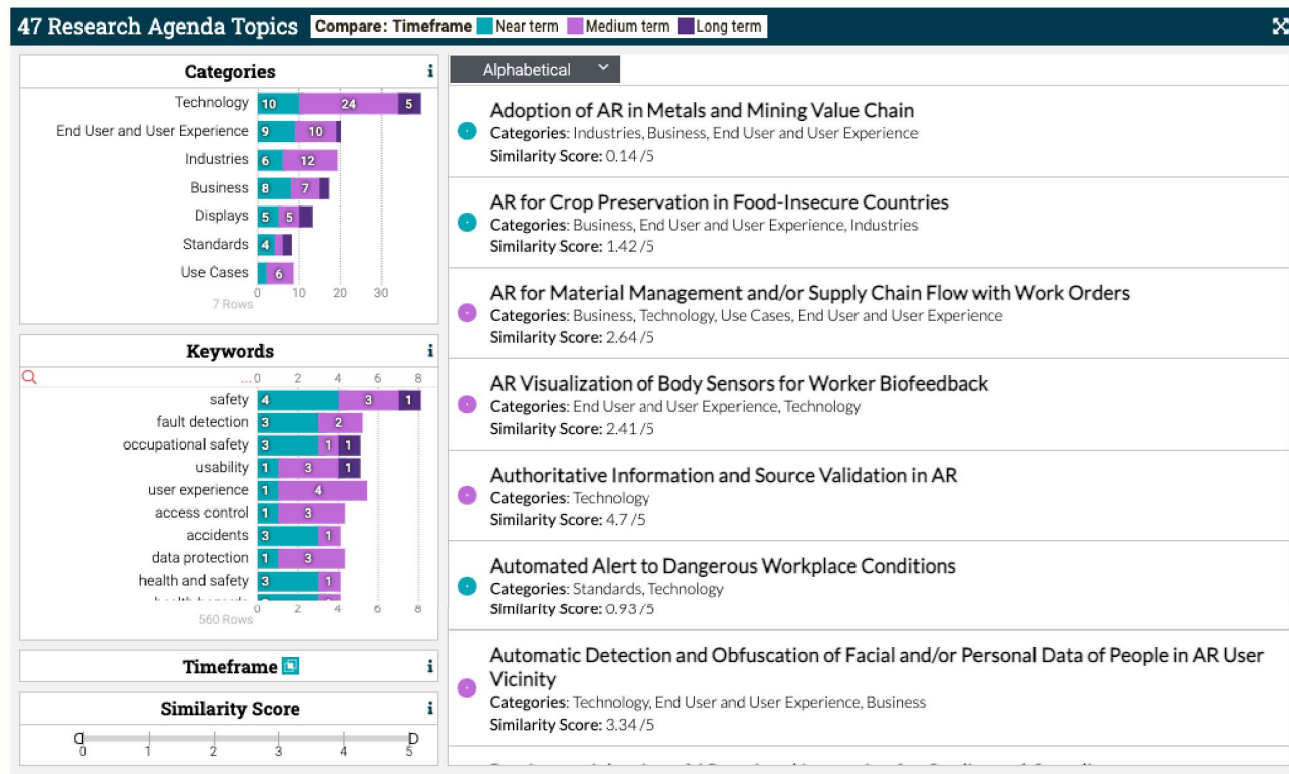


Figure 2: Screenshot of Keshif dashboard designed to help AREA members explore and discover potential research topics.

Before using TF-IDF vectorization to the publication dataset, we removed all “stopwords” from each record’s abstract. We leverage the existing stopwords corpus from the NLTK Python library [1]. We then “hyphenate” each individual keyword in all records and concatenate each record’s hyphenated keywords to the truncated abstract. In doing so, all subsequent data processing stages treat each multi-word keyword as a single word.

3.9 Judging Record Relevance to AREA Members

With each record’s abstract and keywords vectorized through TF-IDF, we can leverage machine learning techniques. We first aimed to semi-automatically classify which publications would be of most interest to AREA membership. To do so, we employed an active learning approach, wherein a set of experts labeled about 20% of all publication records as either relevant or not. Then, a Stochastic Gradient Descent Classifier [3] was used to predict the relevance of the remaining unlabeled data. Paper relevance was based on the TF-IDF vectorization of the cleaned abstract alone over 1000 iterations then leveraged the keywords to predict the rest.

3.10 Judging Relatedness of Records

With the TF-IDF vectorization of the concatenation of the clean abstract and hyphenated keywords, we measured the cosine similarity of each record’s corpus against all other records. In doing so, we identified the top-20 most related (or similar) records per record.

3.11 Ranking Research Opportunities

After the research topics were written up, we used the same cosine similarity-based technique to quantify the similarity of the research topic descriptions against all records in the publication dataset⁵.

⁵More information about the scoring method is found at this link.

3.12 Preparing Research Topic Descriptions

Using a template, we developed prose descriptions of topics which were identified as having low incidence of coverage in the publication dataset and, yet, represent significant barriers to enterprise AR adoption. Five authors with knowledge of diverse, relevant domains and research, contributed. In addition to explaining the central research topic and assigning it to one or more of the seven categories, the description offers related research topics, reasons why AREA members seek results from the proposed research, and possible methodologies to approach the research question. The description includes the publications which have the highest proximity to the research topic, based on bag of words analysis.

3.13 Validating Research Opportunities via Experts

To ensure that the topics identified were reflective of concerns of the industrial AR community and AREA members, the topic descriptions were reviewed and comments provided by 15 subject matter experts. The experts rated how directly or significantly the proposed research topics would impact the challenges they face when implementing AR in the workplace. They also provided suggestions to increase the descriptions’ clarity, applicability, and relevance.

3.14 Publishing with Interactive Dashboard

After the experts’ input was incorporated, updated topic descriptions were published using a WordPress page template on the AREA web site⁶. To increase usability of the research agenda and findability of topics, an interactive dashboard was developed using Keshif⁷, as shown in Fig. 2. The dashboard permits users to explore enterprise

⁶The dashboard can be found at <https://thearea.org/area-research/area-research-agenda>.

⁷<https://keshif.me/>

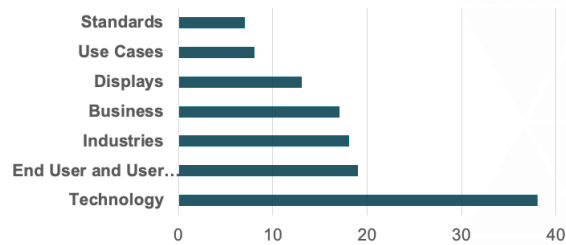


Figure 3: Number of research topics tagged with each category.

AR research topics using categories and keywords of interest. The titles of research topics matching the selected parameters appear on the right side of the dashboard. The topic titles presented are ranked in order of similarity between a topic's descriptive text and prior published research on enterprise AR. Clicking on the title opens the research topic description page in a new browser window.

4 RESULTS

The lack of a coherent enterprise AR research agenda, based on objective, quantitative assessment of existing knowledge as published in the peer reviewed journals, results in many industrial and academic research projects focusing on topics that do not address the needs of large enterprise AR customers seeking to adopt the technology. This project provides members of the AR research community and those who fund future research a set of topics of interest to members of the largest industry association, the AR for Enterprise Alliance. Using the dashboard and studying the project descriptions permits stakeholders to prioritize for further study topics likely to address or meet customer and ecosystem requirements.

4.1 Research Topics

The enterprise AR research topic descriptions are distributed in seven categories (each topic could be assigned multiple categories). Figure 3 shows that most of the topics are attributed to the technology category. Ten of the topics pertain to specific industry domains, including aviation, automotive, oil and gas, chemicals, metals and mining, logistics and agriculture. Seven topics focus specifically on user interaction and improvements to the AR user experience that will reduce barriers to widespread adoption. Seven topics identify specific areas where development of standards will contribute significantly to adoption as a result of increased interoperability and integration of AR in enterprise IT systems. Use cases are the primary focus another seven topics. The use cases of high importance are AR-assisted inspection, navigation, safety, simulation, and use of AR to perform services and inform citizens in urban environments. Business topics include research on integration with existing IT, measuring business and human performance and policies.

4.2 Gaps & Opportunities

Bucketing terms to groups helped identify several limitations.

- Many of the Engineering Village terms are not in common use in the field of computer science (in general) or in the specific computing segments of concern to this project. For example, locks (fasteners), multimedia computing, microcomputers, groupware, and computer mediated communication. One of the terms is "earth." Meanwhile, the term "technology," which lacks all specificity and applies to many aspects of computing, is also one of the low-level terms.
- Some terms currently in use by those working in the field of Augmented Reality (and Mixed Reality, more generally) do not exist in the list of terms generated or used by the system. For example, "anchor", "trackable", and "trigger" are widely

used for describing real world conditions in relation to AR authoring and delivery systems. Other terms that are widely used in computing that are not found in the low-level list include "point clouds" (which could also have dense point clouds, sparse point clouds), and, even more basic, "file format." At the intersection of these, there could be included the names of point cloud file formats (for example, E57).

- For searching for articles pertaining to spatial computing research, commonly used terms such as "Cartesian coordinates" and "Euler angles" are entirely absent in the low-level list.
- Some terms are in the list, however, lack specificity and richness. For example, drones are an important source of real world data and there are many subtleties which could be detected. However, the term "drones" and "autonomous aerial vehicles" each appear once. "Metadata" also only appears once, yet metadata are pervasive across relevant research.
- Terms with which one could search for features of AR display technologies are also lacking.
- Finally, many terms were absent at the time of this research, but would be valuable in the preparation of the AREA Research Agenda. They include terms about geographies (e.g., regional or nation state names), the business aspects of introducing or deploying AR, standardization (e.g., the names of standards development organizations like IEEE, Khronos Group and Open Geospatial Consortium) and about the users of AR systems.

Despite these shortcomings, enterprise AR research is continuing and new peer-reviewed journal articles and conference proceedings are published and indexed by the services we used on an ongoing basis. To reflect the most current state of the art in the field of enterprise AR, we will update and expand the enterprise AR research agenda topics. If and when identified research topics receive attention and sufficient new results are published in the peer-reviewed literature, a methodology for determining when a research topic no longer meets the research agenda criteria will need to be developed.

5 CONCLUSION

Due to the absence of a coherent and objective research agenda, research on enterprise AR topics has focused deeply on some areas with little new results, and completely ignored or neglected other topics. The AREA Enterprise AR Research Agenda described in this paper addresses the issue by providing nearly 50 research topics of high potential interest and impact that have not, to date, received attention and on which researchers are encouraged to focus.

The research agenda described in this paper is a living resource which is being updated by the authors. Feedback and new topics may be contributed by the enterprise AR research community by contacting the authors. The authors also invite the community to assist in the monitoring of new research on topics identified in 2021.

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